UTKARSH SHARMA

Software Systems Builder & Data Science Undergraduate

B.Tech in Computer Science & Engineering (Specialization: Data Science) · NIET (2025–2029)

GitHub: github.com/shutkarshX
LinkedIn: in/utkarsh-sharma
Production App: tein.onrender.com
Location: Greater Noida, Uttar Pradesh, India

2,600+	1,830+	190+	Production
TOTAL COMMITS	CONTRIBUTIONS	PASSING TESTS	RENDER CLOUD LIVE

01 / PROFILE — BUILDER ENGINEERING PHILOSOPHY

2nd-year Computer Science undergraduate at NIET specializing in Data Science and systems engineering. Hands-on builder who learns by turning ideas and technical questions into concrete prototypes, production web platforms, and native desktop systems. Backed by verified open-source delivery, automated testing discipline (190+ passing unit tests), and structured data science internship execution. AI is used thoughtfully as a workflow accelerator while maintaining rigorous personal ownership of system architecture, debugging, test suites, and production deployment.

02 / CAPABILITIES — TECH STACK & SYSTEMS ARCHITECTURE

LANGUAGES

`Python` `Rust` `TypeScript` `JavaScript` `C (C99)` `Java` `SQL` `R`

FRAMEWORKS & DATA

`FastAPI` `Flask` `React` `Vite` `Tauri 2` `Pandas` `Scikit-Learn` `Seaborn`

SYSTEMS & ARCHITECTURE

`Docker` `PWA / Service Workers` `Win32 API` `REST APIs` `Asyncio` `Leaflet GIS`

DEVOPS & VERIFICATION

`Git / GitHub Actions` `Render Cloud` `Pytest (Automated)` `Chrome MV3` `Linux CLI`

03 / TRACK RECORD — PRODUCTION PLATFORMS & INTELLIGENT SYSTEMS

TEIN — Attendance Intelligence & Campus Hub Platform

PRODUCTION

tein.onrender.com ↗ Showcase Repo ↗

Full-stack, mobile-first university platform deployed live on Render solving attendance compliance and student society organization across semester cycles.

- **Bunk & Recovery Logic:** Implemented decision algorithms calculating exact safe lecture skip thresholds and mandatory recovery chains to safeguard the institutional 75% limit.
- **What-If Simulator:** Forward-looking planner allowing students to simulate hypothetical leave scenarios across future dates before taking time off.
- **Campus Community Hub:** Built an integrated directory indexing profiles, domains, and event broadcasts for 20+ university societies and technical clubs.
- **Offline PWA Architecture:** Implemented service worker caching (`/sw.js`) enabling fast, reliable operation over slow campus networks.
- **Production Deployment:** Containerized with Docker, served via Gunicorn WSGI on Render Cloud, backed by 25 automated unit tests.

`Python 3` `Flask` `Docker` `PWA` `Gunicorn` `Render Cloud` `25 Unit Tests`

RESQ-AI — Intelligent Disaster Response Command Center

github.com/shutkarshX/resq-ai ↗

Emergency operations command platform designed for municipal crisis management, real-time rescue prioritization, and geospatial team routing during flood disasters.

- **AI Incident Triage:** Integrated Google Gemini AI to parse citizen emergency dispatches, extract geospatial coordinates, and classify urgency severity into prioritized dispatch tickets.
- **Geospatial Command Map:** Rendered interactive Leaflet maps plotting active rescue convoys, evacuation shelters, and live incident clusters.
- **Async REST API:** High-throughput FastAPI backend with typed Pydantic request models and interactive OpenAPI/Swagger documentation.
- **Real-Time Operations:** Engineered asynchronous endpoint handlers ensuring non-blocking dispatch processing under heavy simulated incident load.

`FastAPI` `React` `Gemini AI` `Leaflet Maps` `Python Async` `Swagger`

Layer-Flow — Transparent Screen Drawing & Desktop Canvas

github.com/shutkarshX/Layer-Flow ↗

Native, lightweight Windows desktop application allowing presenters and educators to draw, highlight, and annotate directly over any active screen in real time.

- **Rust & Tauri 2 Architecture:** Sub-50MB memory footprint and zero UI latency, completely avoiding heavy Chromium wrapper overhead.
- **Win32 Hook Integration:** Implemented low-level OS hooks for global keyboard hotkeys and click-through pass-through state toggling.
- **Multi-Monitor Support:** Handled DPI scaling and multi-display coordinate translation for consistent multi-screen pen and brush strokes.

`Rust` `Tauri 2` `React` `Win32 API` `Desktop Systems`

Speedo — Coursework Progression & Adaptive Video Controller

github.com/shutkarshX/Speedo ↗

Manifest V3 Chrome extension designed to automate learning progression, bypass player speed restrictions, and safeguard coursework progression.

- **CSP-Safe Injection:** Implemented a Main World execution bridge adhering to strict Manifest V3 Content Security Policies without security violations.
- **Modal Dismissal Guard:** Automatically intercepts and handles idle detection popups to maintain uninterrupted video progression.
- **Playback State Persistence:** Preserved adaptive speeds and volume states across playlist segments using Chrome storage APIs.

`JavaScript` `Chrome Extension MV3` `DOM Automation` `CSP Safe`

Integrated National Citizen Services & CPU Scheduling Lab

github.com/shutkarshX/...-Platform ↗

Systems engineering lab implementing and verifying fundamental process scheduling algorithms with an exhaustive automated test suite.

- **Core Scheduling Algorithms:** Implemented First-Come-First-Served (FCFS), Shortest Job First (SJF Preemptive & Non-Preemptive), Priority Scheduling, and Round Robin.
- **166 Automated Tests:** Rigorous test suite validating turnaround time, process starvation resistance, context switching, and CPU utilization.
- **Gantt Timeline Visualization:** Generated precise execution schedules and process state transition graphs for performance verification.

`Python` `Operating Systems` `CPU Scheduling` `166 Automated Tests`

04 / INDUSTRY EXPERIENCE — APPLIED MACHINE LEARNING & DATA SCIENCE

Data Science Internship Track — Prodigy InfoTech

`INTERNSHIP · REMOTE` Sep 2026 — Oct 2026 github.com/shutkarshX/Prodigy-Infotech ↗

Remote data science internship completing a structured 5-task spectrum across exploratory data analysis, predictive modeling, and NLP sentiment mining.

- **Predictive Classification:** Trained and hyperparameter-tuned decision tree classifiers on customer demographic datasets to model subscription propensity; evaluated precision, recall, and ROC-AUC.
- **EDA & Sentiment Mining:** Analyzed demographic distributions, Titanic survival factors, and mined sentiment patterns from social media text feeds using Pandas and Seaborn.
- **Accident Pattern Analysis:** Mapped traffic accident trends against road conditions, weather, and time of day to identify hotspot risk factors.

`Python` `Pandas` `Scikit-Learn` `Seaborn` `EDA` `5 Milestones`

05 / EDUCATION — ACADEMIC CREDENTIALS

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Aug 2025 — Apr 2029

Noida Institute of Engineering & Technology (NIET) · Greater Noida, India

Specialization: Data Science & Engineering Systems.
Core Coursework: Operating Systems, Data Structures & Algorithms, Object-Oriented Software Design (Java), Database Management Systems (DBMS), Systems Programming, Computer Networks, and Statistical Data Analysis.
Campus Initiative: Cataloged and connected 20+ active student societies and technical clubs across the university via TEIN platform.